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Méthode pour évaluer la stabilité à l'oxydation des huiles isolantes

Method for assessing the oxidation stability of insulating oils



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METHOD FOR ASSESSING THE OXIDATION STABILITY OF INSULATING OILS

FOREWORD

- 1) The formal decisions or agreements of the I.E.C. on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
- 2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
- 3) In order to promote this international unification, the I.E.C. expresses the wish that all National Committees having as yet no national rules, when preparing such rules, should use the I.E.C. recommendations as the fundamental basis for these rules in so far as national conditions will permit.
- 4) The desirability is recognized of extending international agreement on these matters through an endeavour to harmonize national standardization rules with these recommendations in so far as national conditions will permit. The National Committees pledge their influence towards that end.

PREFACE

This recommendation has been prepared by Technical Committee No. 10, Insulating Oils.

Following the issue of the first edition in 1955, which had the status of a report on the progress made in the work of preparing a recommendation, studies were continued and a draft revision was discussed at a meeting held at Interlaken in 1961. As a result of this meeting, a draft was submitted to the National Committees for approval under the Six Months' Rule in November 1961.

The following countries voted explicitly in favour of publication:

Austria	Netherlands
Belgium	Norway
Canada	Portugal
Czechoslovakia	Romania
Denmark	Sweden
Finland	Switzerland
France	Turkey
Germany	Union of Soviet Socialist Republics
Hungary	United Kingdom
Israel	United States of America
Italy	

METHOD FOR ASSESSING THE OXIDATION STABILITY OF INSULATING OILS

1. Introduction

The description of this method specifies the conditions for bringing about a series of successive physical and physico-chemical reactions of a much more complex nature than measurement of a simple physical property such as density, viscosity or even the fracture of a metal sample under tension.

It is, therefore, essential that such a test should be carried out under the supervision of an expert, both as regards the manner in which it is done and the interpretation of results.

These are relatively easy in the case of an already known oil, but may be much more difficult where a new product is concerned. It is known, for example, that certain unrefined oils now contain more sulphur than formerly. It may therefore be feared, when examining an unknown oil whose sulphurous constituents are liable to contaminate the metallic catalyst, that abnormal results will be obtained even if every care has been taken to utilise the correct apparatus and to entrust the operations to experienced staff.

The conclusion to be drawn from the above is that special cases will have to be studied separately. The test has been primarily designed for the examination of pure refined petroleum distillates. Its extension to other products is an extrapolation.

The method is, in principle, applicable to inhibited oils. In that case, it may be necessary to take extra precautions as the test necessarily takes much longer.

2. Principle

The oil to be tested, through which a stream of oxygen is bubbled, is maintained for a carefully measured period at a temperature of 100°C in the presence of metallic copper. The degree of oxidation is estimated by determining the amount of sludge and the neutralization value.

3. Test vessel

The test vessel is a round-bottomed borosilicate glass or neutral glass test-tube of the following dimensions:

External diameter:	25 mm
Internal diameter:	22.5 ± 0.5 mm
Wall thickness:	1.25 mm
Length:	150 ± 2 mm
Volume:	approximately 50 ml

The test tube is covered with a round watch-glass type plate of borosilicate glass or of neutral glass 40 mm in diameter and 2 mm thick having a hole, 7 mm in diameter, for the oxygen lead-in tube. The convex side of the plate is facing downwards.

4. Quantity of oil

The quantity of oil shall be $25 \pm 0.1\text{g}$.

The oil, mixed by stirring, is first filtered through a sintered glass filter, identical with that described under Sub-clause 12.1).

5. Oxygen lead-in tube

The oxygen lead in tube shall be of borosilicate glass or neutral glass, and shall have the following dimensions:

Internal diameter: 3 mm minimum

External diameter: 4.5 mm maximum.

The depth of the oxygen lead-in tubes within the oil is set by their bevelled ends, which touch the bottoms of the test vessels. The plane of their bevel forms an angle of 30° with the axis of the tube.

6. Reagents

a) Oxygen:

Obtained from liquid air
minimum purity: 99.4%.

b) Normal heptane:

Density at 20°C : 0.6836-0.6839

Refractive index at 20°C : 1.3876-1.3879

Solidification temperature: below or equal to -90.72°C .

Distillation:

50% distills between 98.38 and 98.48°C .

A maximum temperature rise of 0.20°C between 20 and 80% recovery *.

c) Indicator solution:

A solution of 2 g of alkali blue 6 B in 100 ml of pure ethyl alcohol.

d) Titration solution:

0.1 N alcoholic potassium hydroxide solution.

7. Oxygen supply

The oxygen shall be thoroughly dried by passing it through concentrated sulphuric acid and over lime or soda lime, or by any other method of equal or greater efficacy.

A 10 litre flask, acting as a surge vessel, smoothes the oxygen flow, excess of which bubbles through a mineral oil contained in a test tube.

Flow rate: 1 litre/hour 0.1 litre.

The method used to check the flow rate is left to the discretion of the operator.

8. Methods of cleaning the test vessels

The test tube shall be chemically cleaned. A method of cleaning which has been found satisfactory is to wash with acetone, distilled water, then with concentrated sulphuric acid which is removed by washing, first with tap water, then with distilled water.

The reagents used must be of chemically pure grade.

The apparatus is dried in an air oven at $105-110^\circ\text{C}$ for at least 3 hours, and then allowed to cool to room temperature in a desiccator in which the test-tubes are kept until they are used.

* Normal heptane as described above is exactly the same as that used as a reference fuel in testing the octane index of internal combustion engine fuels

9. Heating arrangements

The test temperature for the oil is 100°C with a tolerance of $\pm 0.5^\circ\text{C}$.

The temperature of the thermostatic block required to maintain the oil at 100°C governs the temperature of the air above the oil surface. The temperature just below the cover of the test-tube shall be not less than 70°C.

A wrought aluminium block provided with holes to accommodate the test-tubes, described under Clause 3 above, is used for heating. Provision is made in the block for at least four test-tubes, and its temperature is maintained constant by the use of a contact thermometer. The test-tubes are inserted as far as possible into the holes. If they cannot be completely inserted, short metal collars passing through the insulating cover and surrounding each test-tube will ensure heating over the full length of the tube.

10. Catalyst

The metallic copper used as oxidation catalyst consists of a wire of non-annealed commercial electrolytic copper of diameter between 1.00 and 1.02 mm. It is prepared as follows:

- a) Immediately before use, the copper wire is polished with No. 00 emery cloth or an equivalent abrasive cloth (see Appendix A). All traces of abrasive are removed with a lintless filter paper and then with a dry, lintless cloth.
- b) A piece of the polished wire 30.5 cm (± 0.1 cm) long is rolled into a spiral of approximately 2.2 cm external diameter and 5 cm long.
- c) The spiral is thoroughly cleaned by dipping it into chemically pure ethyl ether, dried in air and immediately introduced into the test vessel.
- d) To avoid contamination the prepared copper must be handled only with tweezers. The copper wire shall not be re-used.

11. Duration of test

The test-tube being introduced into the apparatus previously brought to operational temperature, the duration of the test is 164 hours, this being reckoned from the time when the oil reaches a temperature of 100°C and when the flow of oxygen has been adjusted to the prescribed rate of 1 litre/hour. The time taken to adjust the oxygen flow shall not exceed 10 minutes.

The measurements of sludge and neutralisation value, after 48 and 96 hours, are carried out only when it is desired to plot an ageing curve.

12. Determinations on the oxidized oil

12.1 Sludge formation

The sludge shall be precipitated by adhering strictly to the procedure described below:

The sample of 25 g of artificially aged oil is cooled in the dark for 1 hour, and is then poured into an Erlenmeyer flask of 500 ml capacity, fitted with a ground glass stopper. 300 ml normal heptane is then used to recover, by swilling, the oil adhering to the test-tube, copper spiral and oxygen lead-in tube; the heptane is then added to the oil in the flask.

The mixture is then allowed to stand in the dark for 24 hours, at a temperature of $20 \pm 2^\circ\text{C}$, before filtering through a glass filter, previously dried to constant weight. The maximum diameter of the pores of the glass filter shall be between 5 and 15 microns, and is determined in accordance with the method described in Appendix B.

At the start of filtering only a small pressure drop should be used, to prevent the sludge passing through the filter. Cloudy filtrates should be passed through a second time.

All traces of oil shall be carefully removed by repeated washing of the sludge with normal heptane. The total volume of the normal heptane used for the washing of the sludge shall be 150 ml. The filter containing the sludge is dried at 110°C to constant weight.

Sludge adhering to the catalyst, to the test-tube, and to the oxygen lead-in tube is transferred, by dissolving it in small quantities of chloroform (a total of 30 ml), to a tared porcelain vessel. It is then dried at 110°C after the evaporation of the chloroform, to constant weight. The weight of the residue is added to that of the sludge obtained by precipitation with normal heptane.

The total sludge is expressed as a percentage of the initial weight of the oil.

If normal heptane recovered from previous tests is used, it is important that its acidity be checked before re-use. A product which is not neutral must be purified before use.

12.2 Neutralization value

The heptane solution obtained after filtering off the sludge is collected in a 500 ml measuring flask and made up to the mark with n-heptane. Three determinations of the neutralization value are made on 100 ml samples of the heptane oil solution.

Immediately before use, the titrating solution is prepared as follows:

1 to 3 ml of the alkali blue solution and a drop of 0.1 N HCl to sensitize the indicator are added to 100 ml of a mixture of 3 parts of benzene to 2 parts of ethyl alcohol in volume.

The mixture is neutralized by 0.1 N potassium hydroxide to give a red colour comparable to that of a 10% solution of cobalt nitrate $\text{Co}(\text{NO}_3)_2 \cdot 6 \text{H}_2\text{O}$. This colour shall persist for at least 15 seconds.

The same volume of neutralized solvent is added, while stirring, to 100 ml of the heptane solution.

This solution is then titrated with 0.1 N alcoholic potassium hydroxide at a temperature not exceeding 25°C.

The neutralization value shall be calculated according to the following formula:

$$\text{N.V.} = \frac{(n_2 - n_1) 5.61}{5}$$

where:

n_2 = the number of millilitres of 0.1 N alcoholic potassium hydroxide solution necessary to neutralize the normal heptane-oil solution.

n_1 = the number of millilitres of 0.1 N alcoholic potassium hydroxide solution necessary to neutralize 100 ml of normal heptane to which have been added 100 ml of the titrating solution.

13 Referee tests

Taking into account the accuracy that can normally be expected from the basic measurements employed during the oxidation stability test, the following requirements apply to referee tests:

a Number of samples

Referee tests shall be carried out simultaneously in triplicate.

b Weight of sludge

If the spread between the three weights in relation to the arithmetical mean does not exceed the maximum value shown in the table below, then the test is valid.

The sludge content is the arithmetical mean of the three tests expressed as a percentage.

Weight of sludge found (mean) mg	Max. permissible spread of results mg
up to 10	2
up to 20	4
up to 40	6
up to 50	8
up to 100	10
above 100	20

c) *Acidity*

arithmé-

If the maximum spread between the three results does not exceed 40% of their arithmetic mean, then the test is valid and this mean is the result.

APPENDIX A

GRAIN DIMENSIONS FOR A No. 00 ABRASIVE CLOTH

The emery grains used in the manufacture of a No. 00 abrasive cloth shall pass completely through a 67 mesh/cm sieve (170 mesh/inch).

The mesh characteristics are defined by the following dimensions:

Aperture width: 0.0089 cm (0.0035 in)

Wire diameter: 0.0061 cm (0.0024 in)

APPENDIX B

DETERMINATION OF MAXIMUM DIAMETER OF THE PORES OF A GLASS FILTER

It is not easy to determine the mean diameter of the pores of a glass filter.

The determination of the maximum diameter of the pores provides a valuable indication of the retaining capacity of the filter, that is the maximum diameter of a spherical particle that can pass through it.

The maximum diameter of the pores is determined by measuring the air pressure necessary to cause the first air bubble to pass through the filter moistened by the liquid to be tested.

The following relation is used

$$D = \frac{30}{P}$$

in which

D diameter, in microns, of the pore;

surface tension of the test liquid in dynes centimetre;

P observed pressure, in millimetres of mercury

In order to check a filter having a maximum pore diameter of 8 to 18 microns, either water or carbon tetrachloride is used as a test liquid.

The washed filter is completely immersed in the test liquid of which a film of several millimetres thickness will remain on the upper surface of the filter attached to the apparatus (Figure 1, page 19).

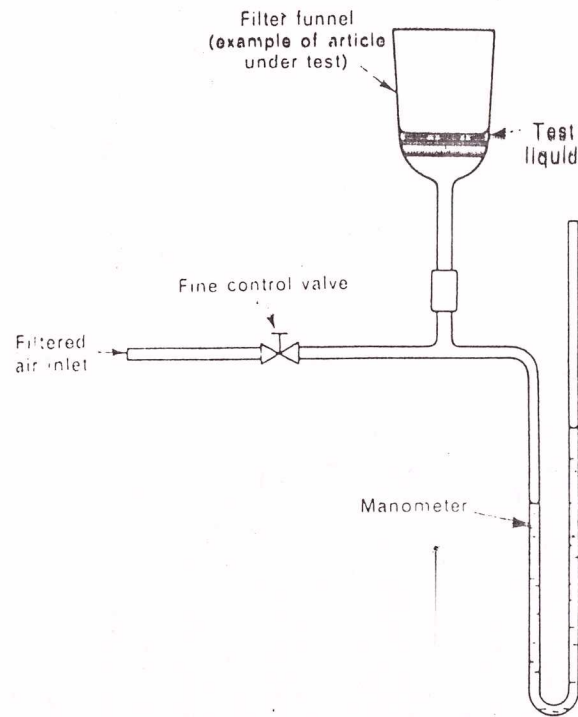


FIGURE 1

A uniformly increasing pressure is applied to the filter until one or two bubbles of air appear in the liquid above the filter. The manometer reading enables the maximum diameter of the pores to be calculated. This manometer can be directly calibrated in pore diameters.

For a filter whose maximum pore diameter is between 5 and 15 microns, the manometer reading in millimetres of mercury will be 435-145 for water and 160-55 for carbon tetrachloride.